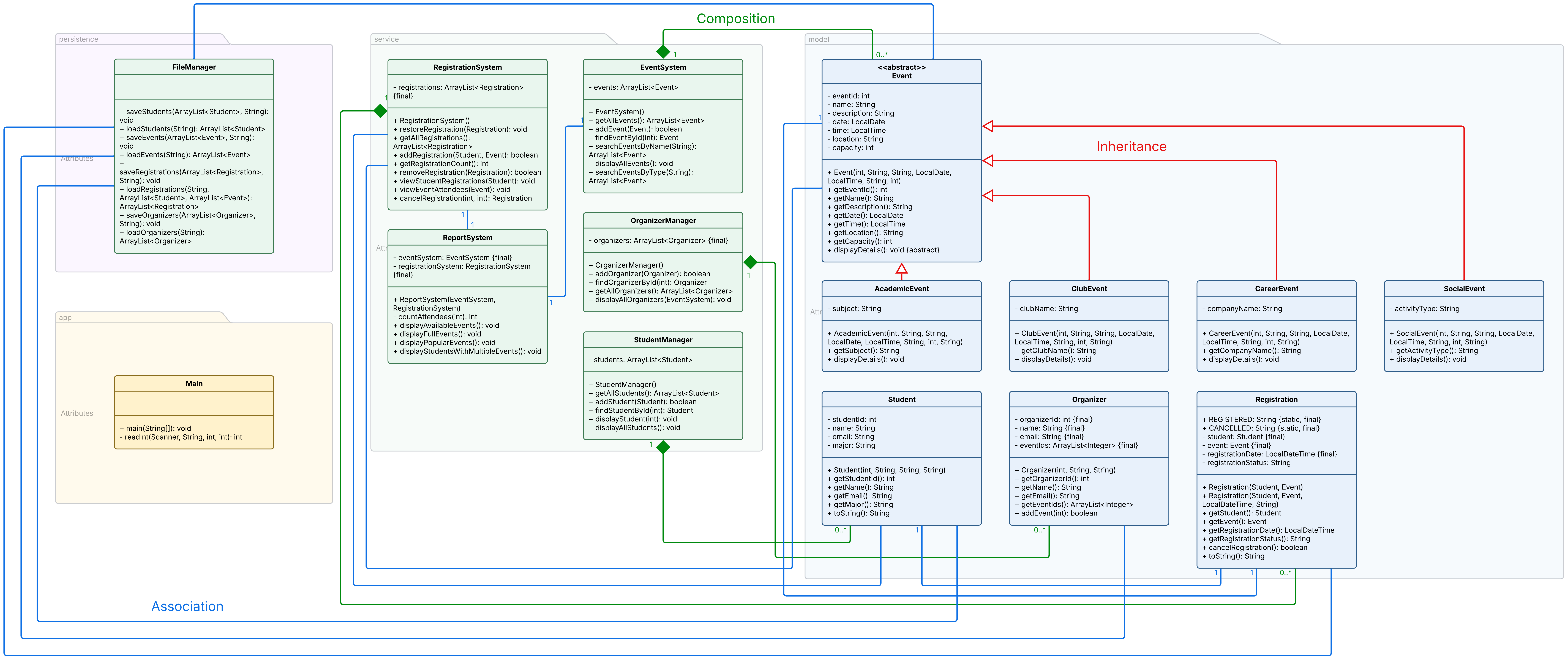

FileManager

+ saveStudents(ArrayList<Student>, String): void
+ loadStudents(String): ArrayList<Student>
+ saveEvents(ArrayList<Event>, String): void
+ loadEvents(String): ArrayList<Event>
+ saveRegistrations(ArrayList<Registration>, String): void
+ loadRegistrations(String, ArrayList<Student>, ArrayList<Event>): ArrayList<Registration>
+ saveOrganizers(ArrayList<Organizer>, String): void
+ loadOrganizers(String): ArrayList<Organizer>

Main

+ main(String[]): void
- readInt(Scanner, String, int, int): int

RegistrationSystem

- registrations: ArrayList<Registration> {final}

+ RegistrationSystem()
+ restoreRegistration(Registration): void
+ getAllRegistrations(): ArrayList<Registration>
+ addRegistration(Student, Event): boolean
+ getRegistrationCount(): int
+ removeRegistration(Registration): boolean
+ viewStudentRegistrations(Student): void
+ viewEventAttendees(Event): void
+ cancelRegistration(int, int): Registration

ReportSystem

- eventSystem: EventSystem {final}
- registrationSystem: RegistrationSystem {final}

+ ReportSystem(EventSystem, RegistrationSystem)
- countAttendees(int): int
+ displayAvailableEvents(): void
+ displayFullEvents(): void
+ displayPopularEvents(): void
+ displayStudentsWithMultipleEvents(): void

EventSystem

- events: ArrayList<Event>

+ EventSystem()
+ getAllEvents(): ArrayList<Event>
+ addEvent(Event): boolean
+ findEventById(int): Event
+ searchEventsByName(String): ArrayList<Event>
+ displayAllEvents(): void
+ searchEventsByType(String): ArrayList<Event>

OrganizerManager

- organizers: ArrayList<Organizer> {final}

+ OrganizerManager()
+ addOrganizer(Organizer): boolean
+ findOrganizerById(int): Organizer
+ getAllOrganizers(): ArrayList<Organizer>
+ displayAllOrganizers(EventSystem): void

StudentManager

- students: ArrayList<Student>

+ StudentManager()
+ getAllStudents(): ArrayList<Student>
+ addStudent(Student): boolean
+ findStudentById(int): Student
+ displayStudent(int): void
+ displayAllStudents(): void

<<abstract>>
Event

- eventId: int
- name: String
- description: String
- date: LocalDate
- time: LocalTime
- location: String
- capacity: int

+ Event(int, String, String, LocalDate, LocalTime, String, int)
+ getEventId(): int
+ getName(): String
+ getDescription(): String
+ getDate(): LocalDate
+ getTime(): LocalTime
+ getLocation(): String
+ getCapacity(): int
+ displayDetails(): void {abstract}

AcademicEvent

- subject: String

+ AcademicEvent(int, String, String, LocalDate, LocalTime, String, int, String)
+ getSubject(): String
+ displayDetails(): void

ClubEvent

- clubName: String

+ ClubEvent(int, String, String, LocalDate, LocalTime, String, int, String)
+ getClubName(): String
+ displayDetails(): void

CareerEvent

- companyName: String

+ CareerEvent(int, String, String, LocalDate, LocalTime, String, int, String)
+ getCompanyName(): String
+ displayDetails(): void

SocialEvent

- activityType: String

+ SocialEvent(int, String, String, LocalDate, LocalTime, String, int, String)
+ getActivityType(): String
+ displayDetails(): void

Student

- studentId: int
- name: String
- email: String
- major: String

+ Student(int, String, String, String)
+ getStudentId(): int
+ getName(): String
+ getEmail(): String
+ getMajor(): String
+ toString(): String

Organizer

- organizerId: int {final}
- name: String {final}
- email: String {final}
- eventIds: ArrayList<Integer> {final}

+ Organizer(int, String, String)
+ getOrganizerId(): int
+ getName(): String
+ getEmail(): String
+ getEventIds(): ArrayList<Integer>
+ addEvent(int): boolean

Registration

+ REGISTERED: String {static, final}
+ CANCELLED: String {static, final}
- student: Student {final}
- event: Event {final}
- registrationDate: LocalDateTime {final}
- registrationStatus: String

+ Registration(Student, Event)
+ Registration(Student, Event, LocalDateTime, String)
+ getStudent(): Student
+ getEvent(): Event
+ getRegistrationDate(): LocalDateTime
+ getRegistrationStatus(): String
+ cancelRegistration(): boolean
+ toString(): String